WenTrans - A CWS powered ancient-modern Chinese translator

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1 Introduction

Chinese Machine Translation has undergone significant breakthroughs over the past years, from rule-based machine translation (RBMT), to statistical machine translation (SMT), and most recently to Neural Machine Translation (NMT) [1]. In the general sense, the direct application of Chinese Word Segmentation (CWS) has been diluted by applications of neural networks which treat Chinese as a continuous string of tokens given its nature of lacking white spaces. Modern subword tokenization methods such as SentencePiece [2] operate directly on raw character sequences. However, it has been shown that CWS still proves to be crucial in Ancient - Modern Chinese translation, as ancient Chinese displays unique features including but not limited to monosyllabic dominance and flexible word boundaries [3]. In this project, we will build upon the SOTA ancient Chinese embedding and segmentation models [4] and architect a hybrid ancient Chinese segmentation pipeline which will be integrated into a modularized ancient-modern translator framework. We will then evaluate the results using BLEU metrics [5].

Our code is available at https://github.com/KiwiOperator/WenTrans
Two branches are available with their names indicating their respective stage in the pipeline.

2 Datasets

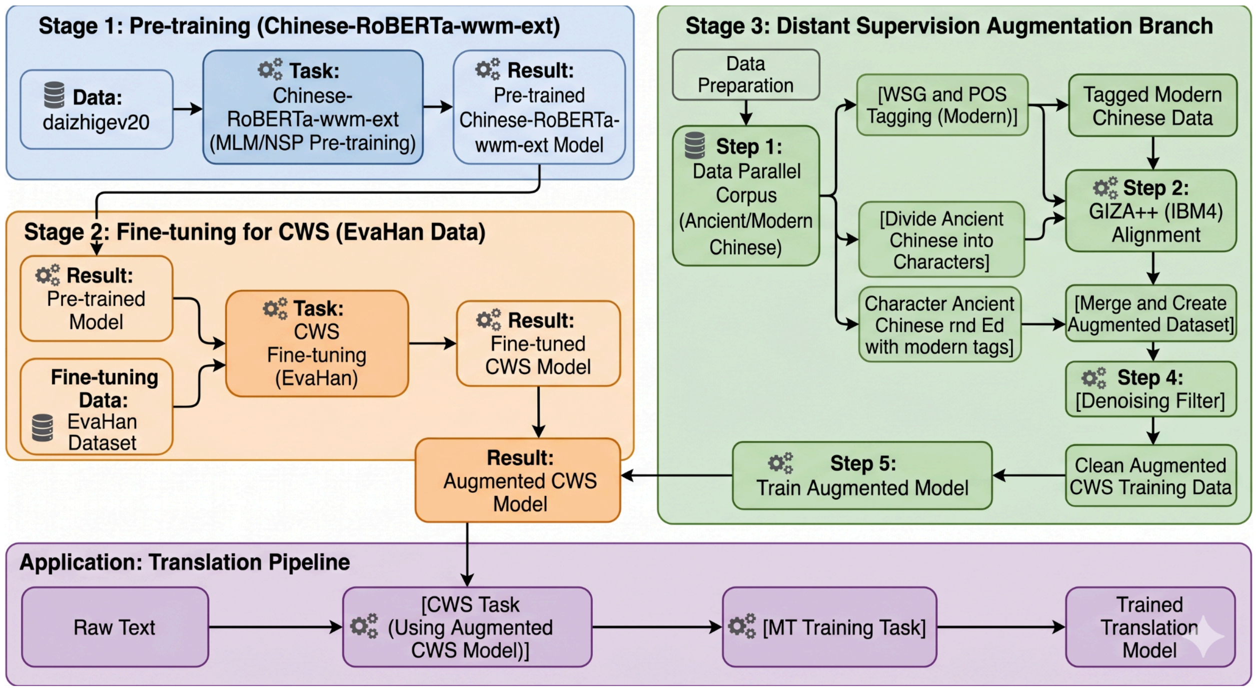
For our CWS model, we will use the EvaHan Zuozhuan corpus, containing 166,142 word tokens as our ground truth word segmentation dataset [6], representing the current standard benchmark for Ancient Chinese segmentation.

Our ancient-modern translator model will use the Erya dataset [7], containing the largest available parallel corpus with 1.24 million bilingual pairs.

3 Methodology

Rather than develop a word segmentation model from scratch, we will fine-tune SikuRoBERTa on the EvaHan Zuozhuan corpus, and use the novel method of Part-of-Speech Tagging using distant supervision to augment this model [8]. This addresses the problem of the scarcity of labelled segmentation data for ancient corpuses by utilizing word boundaries and POS tags for Modern Chinese sentences and projecting them onto their ancient counterparts using word alignment [9]. After obtaining the segmented tokens, we will benchmark our model against the SOTA by calculating F-1 score.

In the second stage of our pipeline, we use a fine-tuned SikuRoBERTa's segmentation as auxiliary input to train the current SOTA translation model Erya, which achieves +12.0 BLEU against GPT-3.5 baselines [10]. We will experiment with injecting segmentation and POS-tagging as boundary tags in the input or adding it as an auxiliary loss during training before benchmarking on the Erya dataset using BLEU.



4 CWS Model

4.1 Pre-training

Building on top of work from [11], we started by pre-training the chinese-roberta-wwm-ext model, a popular transformer-encoder model built specifically for NLP tasks in Chinese. Developed by the joint Laboratory of HIT and iFLYTEK Research, it improves upon the Roberta architecture to fit for Chinese downstream tasks [12]. We perform pre-training using 600 M characters of classical text from DaizhiGeV2 [13] with MLM(Masked Language Modelling) and removing NSP adhering to the pipeline proposed by [11]. After ~14 hours of training on PACE L40S, the encoder reached an eval perplexity of 15.69, a result comparable to first-pass SikuBERT. Specifically on traditional Chinese **historic** literature, its perplexity drops almost 45 times.

Passage	Base RoBERTa-wwm-ext	Ours	SikuRoBERTa	Best
classical_zz (Zuozhuan narrative)	2,257.28	51.45	**24.35**	siku

classical_lunyu (Analects maxim)	**111.31**	442.67	853.89	base
classical_mengzi(Mencius proverb)	**2.16**	16.14	18.36	base
modern_zh	**3.04**	34.24	268.36	base

(Table 1: Pseudo-Perplexity comparison among ext (baseline), WenTrans(our model), and SikuRoBERTa (SOTA)

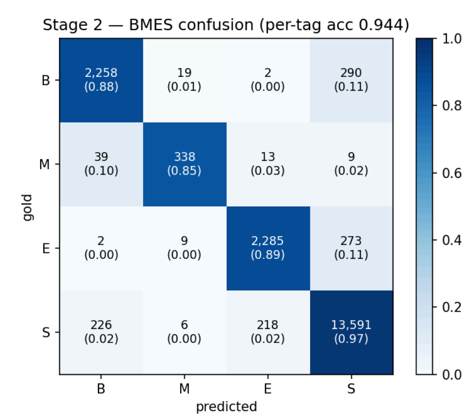
We can see improvements in the model's performance in traditional texts compared with SikuRoBERTa, yet the base model takes the lead in classical literature presumably due to its exposure to these quotes even in a modern training dataset. Here are some qualitative examples showing our model's improved performance in ancient literature.

Sentence (gold)	Base (gold rank)	Ours (gold rank)
生 [MASK] 公而惠公薨 (桓)	63	**6**
學而 [MASK] 習之 (時)	416	**105**
蔓草 [MASK] 不可除 (猶)	1541	4011
天下 [MASK] 公 (為)	60	3656

4.2 CWS Finetuning

We proceed to finetune our model with EvaHan dataset [6], a valuable dataset labelled by experts in ancient Chinese literature (190K characters, 16K sentences) split sentence-wise 8/1/1. On top of our stage-1 encoder, we strip the POS tags, BMES-encode the characters and fine-tune for 10 epochs with AdamW on an AutoModelForTokenClassification head with the four output labels.

Test setup	F1
SikuRoBERTa 2021(self-administered Zuozhuan CWS, 2021 setup)	88.88 %
Feng & Li 2023 baseline (Tina et al.) — EvaHan 2022 Test-A	94.73 %



Feng & Li 2023 M_3 (full pipeline) — EvaHan 2022 Test-A	95.64 %
Ours (Stage2only, 10% holdout, in-domain random)	95.09 %
Ours (Stage 2-equivalent M_2 on EvaHan 2022 Test-A)	**95.71 %**
Test setup	**F1**

Our model's performance is slightly better than the refined model proposed by Feng & Li 2 years after their initial publication, reaching an F-1 score of 95.71%.

4.3 Augmenting the model with distant supervision

There is one key shortage of our model up to this point: Due to limitations in computation capabilities, our pre-training and finetuning datasets revolve around similar topics and literatures of similar writing styles, which makes the robot prone to underperformance in unrelated genres. Taking inspiration from [14], we augmented the model with a larger weakly labelled dataset using distant supervision. By running GIZA++IBM-4 unsupervised aligner over parallel pairs of ancient-modern Chinese corpus, we produced quality data to further improve the model. For each ancient character (we divide ancient characters individually) we pick the alignment with the highest probability and tag it by mapping it through the paper's 29 → 22 modern → ancient POS dictionary. We then merge neighboring ancient characters of the same tag, leaving unaligned characters with tag NULL. Then, denoising operations were performed strictly adhering to the referenced publication: M_1 (our pretrained encoder trained on the weakly-labelled data); M_2 (M_1 continued on the small clean Zuozhuan training set) corrects M_1's noise via the DNN memorisation effect; M_3, the final model, is trained from scratch on a M_2-relabeled version of the weakly-labelled data, combining its scale with M_2's improved label quality.

Stage	Test-A WSG
M_1	82.42
M_2	95.71 (+13.3)
M_3	95.01 (−0.7)

5 Translator with auxiliary inputs

We augment the Erya translation model with soft POS-tag and word-segmentation embeddings from SikuRoBERTa, and use a three-stage training curriculum to fine-tune the model.

5.1 Generating Augmented Dataset

In order to obtain soft POS-tag and word-segmentation embeddings, we run a pretrained SikuRoBERTa-CRF tagger [8] which jointly predicts character-level word-segmentation labels and ~30 POS labels.

5.1.1 Tagger

SikuRoBERTa-CRF is a modification that places two independent classifier heads on top of the encoder's final hidden states:

- classifier_seg → 4 BIES labels (B / I / E / S - Begin, Inside, End, Single)
- classifier_pos → 30 POS labels

We use the checkpoint sikuRoberta_model_crf0.pth trained on the EvaHan ZuoZhuan corpus [6].

```
書/n 日/v ：/w “/w 衛人/n 立/v 晉/nr ”/w ， /w 衆/n 也/y 。
頃公/nr 之/u 嬖人/n 盧蒲就魁/nr 門/v 焉/y 。/w 。
INPUT : 書日衛人立晉。
OUTPUT: 書/n 日/v 衛人/n 立/v 晉/nr 。/w

INPUT : 頃公之嬖人盧蒲就魁門焉。
OUTPUT: 頃公/nr 之/u 嬖人/n 盧蒲就魁/nr 門/v 焉/j 。/w

INPUT : 子日學而時習之不亦說乎。
OUTPUT: 子/nr 日/v 學/v 而/c 時/d 習/v 之/r 不/d 亦/d 說/v 乎/y 。/w
```

Pictured: segmentation and POS outputs from SikuRoBERTa-CRF with Viterbi-decoded hard tags

Rather than store the Viterbi-decoded hard tags, we save the full softmax distribution, since this allows the downstream model to consume soft tag information. This way, a character with ambiguous segmentation has a different signal from a confidently tagged character, which allows the downstream model to attend to confidence as well as tag identity.

5.2 TaggedErya

We use Erya as our base translation model (BART-like, 12-layer encoder, 2-layer decoder, d_model=768, vocab=51271, max_position embeddings=1024) pretrained on the Erya dataset. We use BertTokenizer using inference to tokenize CJK (Chinese character) input.

5.2.1 TaggedErya Wrapper

We add tag conditioning by introducing three small trainable components in our wrapper:

Component	Type	Shape	Init
`E_seg`	`nn.Embedding`	`(num_seg, d_model) = (4, 768)`	`N(0, 0.02)`
`E_pos`	`nn.Embedding`	`(num_pos, d_model) = (30, 768)`	`N(0, 0.02)`
`alpha_seg`, `alpha_pos`	`nn.Parameter` (scalar)	`()`	`0`

During the forward pass we mirror the augmenter's character-level tokenization and construct the encoder inputs explicitly:

$$\mathbf{e}_{\text{tok}} = E_{\text{tok}}(\mathbf{x}) \cdot s_e$$

$$\mathbf{e}_{\text{seg}} = (\mathbf{p}_{\text{seg}} \in \mathbb{R}^{T\times 4}) \cdot E_{\text{seg}}, \qquad \mathbf{e}_{\text{pos}} = (\mathbf{p}_{\text{pos}} \in \mathbb{R}^{T\times 30}) \cdot E_{\text{pos}}$$

$$\mathbf{e}_{\text{in}} = \mathbf{e}_{\text{tok}} \;+\; \tanh(\alpha_{\text{seg}}) \cdot \mathbf{e}_{\text{seg}} \;+\; \tanh(\alpha_{\text{pos}}) \cdot \mathbf{e}_{\text{pos}}$$

We pass in e_in to the BART encoder using the model(inputs_embeds=e_in, …). HuggingFace's BartEncoder.forward skips its internal embed_tokens * embed_scale step if e_in is supplied, but it still adds learned positional embeddings, leaving the rest of the encoder stack unchanged.

5.2.2 Auxiliary KL Distillation Head

We use a linear head (losses.py:KLDistillHead) to project the encoder's last hidden state to (num_seg + num_pos) logits. In order to encourage the encoder's representation to retain tag-recoverable information instead of collapsing under the translation objective, we minimize

$$\mathcal{L}_{\text{KL}} = \frac{1}{|\mathcal{V}|} \sum_{t\in\mathcal{V}} \Big[\mathrm{KL}\big(p^{T}_{\text{seg},t} \,\|\, p^{S}_{\text{seg},t}\big) + \mathrm{KL}\big(p^{T}_{\text{pos},t} \,\|\, p^{S}_{\text{pos},t}\big)\Big]$$

Teachers p^T are SikuRoBERTa softmaxes (from the tagger), students p^S come from the linear head, and V is the set of non-special, non-pad positions with non-zero teacher mass (so denoising batches with zeroed teachers don't contribute).

5.2.3 Parameter count

Stage	Trainable params	Frozen params
1 — adapters only	~75 k (`E_seg` + `E_pos` + 2 scalars + KL head)	~410 M
2 — adapters + encoder	~125 M	~285 M
3 — full model	~410 M	0

5.3 Training

We introduce a three-stage training regime that introduces the new trainable parameters at initialization, and progressively unfreezes parts of the pretrained Erya model that need to consume them.

- **Stage 1:** Find a useful tag-embed subspace
  - Trainable: E_seg, E_pos, α_*, KL head
  - Frozen: encoder, decoder, LM head
- **Stage 2:** Encoder learns to use tag-conditioned input
  - Trainable: add 12-layer encoder
- **Stage 3:** Re-align decoder to new encoder representations
  - Trainable: add decoder, LM head, embeddings

5.4 Experiments

We trained on a single H100 SMX5 80GB for approximately 7 hours.

Stage	Optimizer steps	Wall-clock
Stage 1	8,000	~4 h 30 m (with full-valid passes)
Stage 2	16,000	~1 h 25 m (with `valid_max_batches=200`)
Stage 3	8,000	~1 h 30 m (with `valid_max_batches=200`)

Our best validation cross-entropy from 3.65 to 1.91 to 1.78 from stages 1, 2, and 3. Unfreezing the encoder and training the decoder in stage 2 produced the steep ~50% drop in CE.

We benchmarked on the *hans* test split (Book of Han) of the Erya corpus, obtaining the following results

Variant	BLEU	chrF
Stage 1 best	0.04	1.71
Stage 2 best	27.80	25.18
Stage 3 best, adapter on (`α_seg=+0.41` , `α_pos=−0.78`)	**28.48**	**25.73**
Stage 3 best, adapter neutralised (`α=0`)	22.99	21.43

$$\Delta^{\text{BLEU}}_{\text{adapter}} = 28.48 - 22.99 = +5.49 \quad (+19\% \text{ relative})$$

$$\Delta^{\text{chrF}}_{\text{adapter}} = 25.73 - 21.43 = +4.30 \quad (+17\% \text{ relative})$$

We compare the trained-α row to the α=0 row to isolate the adapter's effect at inference, finding that soft-tag injection contributes ~+5.5 BLEU on the *hans* split. This suggests that the encoder learns to depend on tag-conditioned input embeddings during stages 2 and 3, and that removing them at inference time produces distribution-shifted hidden states that the decoder can not cleanly translate.

BLEU score compared to other SOTA translation models

Model	Book of Han	Ming History	Taiping Guangji	Xu Xiake	Tang History
gpt-3.5-turbo	17.7	21.5	16.8	20.1	20.5
Erya	29.9	37.1	24.1	34.2	34.5
TaggedErya	**28.5**	**36.7**	**22.0**	**33.0**	**35.3**

Our results suggest that TaggedErya is able to hold up against current SOTA translation models, even surpassing Erya on the Tang History split. At 28.5 BLEU, our model is in the same range as published Erya zero-shot.

6 Summary

WenTrans is a small instance of a broader pattern in current NLP: adapting a frozen pretrained backbone with a tiny learned module, applied to Ancient→Modern Chinese translation. We distill a domain-specialist SikuRoBERTa POS/segmentation tagger into ~75k trainable parameters that gate-residual into Erya's encoder, yielding +5.5 BLEU on hans over the same model with the gates zeroed. Extensions include formally integrating our CWS model into the TaggedErya translation pipeline, as its benchmark results seem promising compared to previous work.

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